

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE OUTCOME COVERED

COURSE CODE:CSA1708

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I s.sanjai(192421121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sanjai S

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$ 4)
- Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

[2,2]. An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Let:

F = Patient A has Fever
C = Patient A has Cough
B = Patient A has Breathlessness
Flu = Patient A has Possible Flu
M = Patient A has Possible Measles
P = Patient A has Possible Pneumonia

Rules:

1. Fever AND Cough $\rightarrow$ Possible Flu
 $F \wedge C \rightarrow \text{Flu}$
2. Fever AND Rash $\rightarrow$ Possible Measles
 $F \wedge R \rightarrow M$
3. Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
 $C \wedge B \rightarrow P$

Facts:

F = True
C = True
B = True

Symbol definitions:

$\wedge$ = AND
 $\rightarrow$ = Implies / If-Then
True = The fact is known to be true

(b) Forward Chaining

Forward chaining starts with the **known facts** and applies rules whose conditions are satisfied.

Initial Facts:

F = True, C = True, B = True

Step 1: Apply Rule 1

Rule: **$F \wedge C \rightarrow \text{Flu}$**

Since Fever = True and Cough = True:

Flu = True

Derived: Patient A has Possible Flu.

Step 2: Apply Rule 3

Rule: $C \wedge B \rightarrow P$

Since Cough = True and Breathlessness = True:

P = True

Derived: Patient A has Possible Pneumonia.

Step 3: Apply Rule 2

Rule: $F \wedge R \rightarrow M$

Fever is True, but **Rash is not given.**

Therefore:

Rule 2 cannot be fired.

Final Derived Diagnoses

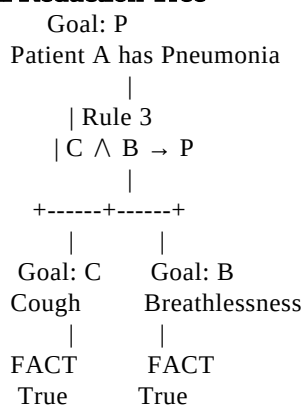
Possible Flu = True
Possible Pneumonia = True
Possible Measles = Not derived

(c) Backward Chaining

Goal: Patient A has Pneumonia (**P**)

Start with the goal and work backward to find the rules that can prove it.

Goal Reduction Tree



Verification

1. Goal is **P (Pneumonia)**.
2. Rule 3 can derive P:
 $C \wedge B \rightarrow P$
3. Check **C (Cough)** → True.
4. Check **B (Breathlessness)** → True.
5. Both required facts are available.
6. Therefore:

P = True

Conclusion

Patient A has Possible Pneumonia according to the given rule-based knowledge system. The goal is successfully verified using **Backward Chaining**.

Case Study 2 — Autonomous Traffic Management Agent (a)

Unification of Rule 1

Rule 1:

$$\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$$

Convert the rule into clause form:

$$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$$

Given facts:

$$\text{Vehicle}(\text{Ambulance}) \quad \text{EmergencyType}(\text{Ambulance})$$

Unification Steps

Step 1: Match

$$\text{Vehicle}(x)$$

with

$$\text{Vehicle}(\text{Ambulance})$$

Therefore,

$$\theta_1 = \{x/\text{Ambulance}\}$$

Step 2: Apply the substitution to the second condition:

$$\text{EmergencyType}(x) \Rightarrow \text{EmergencyType}(\text{Ambulance})$$

This matches the given fact:

$$\text{EmergencyType}(\text{Ambulance})$$

So,

$$\theta_2 = \{x/\text{Ambulance}\}$$

Final MGU

$$\theta = \{x/\text{Ambulance}\}$$

Therefore, Rule 1 produces:

$$\text{ClearPath}(\text{Ambulance})$$

(b) Resolution Proof for Proceed(CarA)

Step 1: Convert relevant rules to CNF

Rule 1:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

Split into two clauses:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$ $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule 3:

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Step 2: Given Facts

$\text{Vehicle}(\text{Ambulance})$ $\text{EmergencyType}(\text{Ambulance})$ $\text{Vehicle}(\text{CarA})$ $\text{Behind}(\text{CarA}, \text{Ambulance})$

Step 3: Derive ClearPath(Ambulance)

Using Rule 1:

$\neg \text{Vehicle}(A) \vee \neg \text{EmergencyType}(A) \vee \text{ClearPath}(A)$

Resolve with:

$\text{Vehicle}(\text{Ambulance})$

and

$\text{EmergencyType}(\text{Ambulance})$

Therefore:

$\text{ClearPath}(\text{Ambulance})$

Step 4: Derive GreenSignal(Ambulance)

Rule 2:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Substitute $x = \text{Ambulance}$:

$\neg \text{ClearPath}(\text{Ambulance}) \vee \text{GreenSignal}(\text{Ambulance})$

Resolve with:

$\text{ClearPath}(\text{Ambulance})$

Therefore:

$\text{GreenSignal}(\text{Ambulance})$

Step 5: Negate the Goal

Goal:

$\text{Proceed}(\text{CarA})$

For resolution by contradiction, add:

$\neg \text{Proceed}(\text{CarA})$

Rule 3:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Substitute:

$x = \text{CarA}, y = \text{Ambulance}$

Giving:

$\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

Resolve with the facts:

$\text{Vehicle}(\text{CarA}) \text{ Behind}(\text{CarA}, \text{Ambulance}) \text{ GreenSignal}(\text{Ambulance})$

We finally obtain:

$\text{Proceed}(\text{CarA})$

Resolve with the negated goal:

$\text{Proceed}(\text{CarA}) \neg \text{Proceed}(\text{CarA})$

Result:

□

where □ represents the **empty clause/contradiction**.

Conclusion

Since the negation of the goal leads to a contradiction:

$\therefore \text{Proceed}(\text{CarA})$ is true

(c) Handling Two Simultaneous Emergency Vehicles Is the current knowledge base sufficient?

No, not completely.

The current rules can handle **multiple vehicles individually**, because Rule 1 and Rule 2 use the variable x :

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

However, the knowledge base does not explicitly contain facts or rules for **two simultaneous emergency vehicles** and their interaction.

Additional Facts Needed

Suppose there are two emergency vehicles:

$\text{EmergencyType}(\text{Ambulance1}) \text{ Vehicle}(\text{Ambulance1}) \text{ EmergencyType}(\text{Ambulance2}) \text{ Vehicle}(\text{Ambulance2})$ If

their positions are relevant, additional facts such as:

$\text{Behind}(\text{CarA}, \text{Ambulance1}) \text{ Behind}(\text{CarB}, \text{Ambulance2})$

are needed.

Additional Rule

A useful rule would be:

$\forall x \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

The existing Rule 1 already handles each emergency vehicle separately.

An additional **conflict-management rule** would be useful, for example:

$\text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge x=y \rightarrow \text{Priority Management}(x,y)$

This allows the system to manage situations where two emergency vehicles require priority simultaneously.

Logical Justification

For each emergency vehicle, the system can independently derive:

$\text{EmergencyType}(\text{Ambulance1}) \rightarrow \text{ClearPath}(\text{Ambulance1}) \rightarrow \text{GreenSignal}(\text{Ambulance1})$ and

EmergencyType(Ambulance2) $\rightarrow$ ClearPath(Ambulance2) $\rightarrow$ GreenSignal(Ambulance2)

But if both vehicles require the **same road or intersection**, the current knowledge base does not specify which vehicle should receive priority or how conflicting green signals should be handled.

Final Conclusion

The current KB can identify multiple emergency vehicles, but needs additional priority/conflict rules and location facts to safely manage simultaneous emergencies.

Case Study 3 — Agricultural Expert Reasoning System

Let:

D = SoilDry

I = IrrigationNeeded

W = CropWheat

A = ApplyDripMethod

R = CropAtRisk

(a) Conversion to CNF

P1: SoilDry $\rightarrow$ IrrigationNeeded

Step 1: Eliminate implication

$$D \rightarrow I \equiv \neg D \vee I$$

So CNF is:

$$\neg D \vee I$$

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Original:

$$(I \wedge W) \rightarrow A$$

Step 1: Eliminate implication

$$\neg(I \wedge W) \vee A$$

Step 2: Move $\neg$ inward using De Morgan's law

$$(\neg I \vee \neg W) \vee A$$

Therefore:

$$\neg I \vee \neg W \vee A$$

This is already in CNF.

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Original:

$$(\neg A \wedge W) \rightarrow R$$

Step 1: Eliminate implication

$$\neg(\neg A \wedge W) \vee R$$

Step 2: Move $\neg$ inward

Using De Morgan's law:

$$(A \vee \neg W) \vee R$$

Therefore:

$$A \vee \neg W \vee R$$

This is already in CNF.

Facts

Given:

SoilDry=True CropWheat=True

Therefore:

D W

Complete CNF Knowledge Base

$$\neg D \vee I \neg I \vee \neg W \vee A \vee \neg W \vee R \vee D \vee W$$

(b) Resolution Refutation to Prove ApplyDripMethod

Goal

A

For resolution refutation, **negate the goal**:

$\neg A$

Add it to the CNF knowledge base.

Resolution Step 1

From P1:

$$\neg D \vee I$$

Fact:

D

Resolve:

I

Therefore:

SoilDry $\Rightarrow$ IrrigationNeeded

Resolution Step 2

Use P2:

$\neg I \vee \neg W \vee A$

We have:

I

Resolve:

$\neg W \vee A$

Resolution Step 3

Use fact:

W

Resolve with:

$\neg W \vee A$

Therefore:

A

So we have derived **ApplyDripMethod**.

Resolution Step 4: Contradiction

We added the negated goal:

$\neg A$

But we derived:

A

Resolving them:

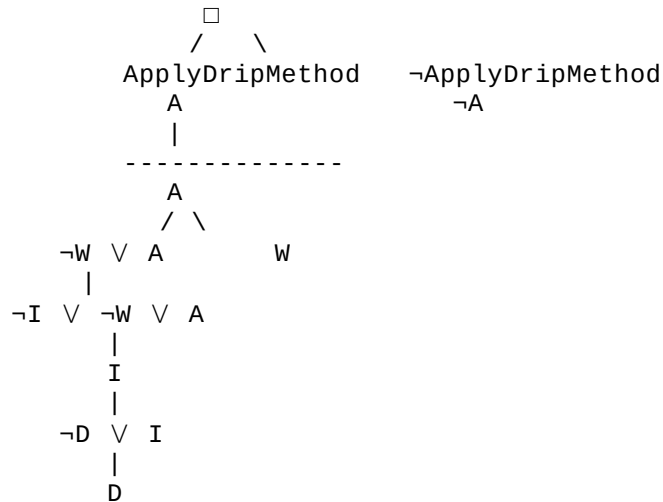
$A, \neg A$

gives:

□

The empty clause means a contradiction has been reached.

Resolution Proof Tree



Conclusion

Since adding **¬ApplyDripMethod** to the KB produces the empty clause:

$KB \models \text{ApplyDripMethod}$

Therefore, **ApplyDripMethod** is logically proven.

(c) Removing the Fact SoilDry

Now remove:

D

from the knowledge base.

The remaining facts are:

W

and the rules remain:

$\neg D \vee I \vee \neg I \vee \neg W \vee A \vee \neg W \vee R$

Step 1: Can we derive IrrigationNeeded?

P1 is:

$D \rightarrow I$

But **D (SoilDry)** is no longer available.

Therefore:

I cannot be derived

Step 2: Can we derive ApplyDripMethod?

P2 requires:

$I \wedge W$

We know:

$W = \text{True}$

but we do **not** know:

$I = \text{True}$

Therefore:

A cannot be derived

Step 3: Does CropAtRisk follow?

P3 is:

$\neg A \wedge W \rightarrow R$

We know:

$W = \text{True}$

However, the absence of a proof for **A** does **not automatically mean**:

$\neg A = \text{True}$

This is important in classical logic.

Therefore, we cannot establish the complete antecedent:

$\neg A \wedge W$

Hence:

R does not follow

Final Comparison:

Situation	Conclusion
SoilDry = True	IrrigationNeeded is derived
SoilDry = True+ CropWheat = True	ApplyDripMethod is derived
SoilDry removed	IrrigationNeeded cannot be derived
SoilDry removed	ApplyDripMethod cannot be derived
SoilDry removed	CropAtRisk does not follow

Final Conclusion

Removing **SoilDry** breaks the reasoning chain:

SoilDry → IrrigationNeeded → ApplyDripMethod

Without **SoilDry**, the system cannot prove **IrrigationNeeded** or **ApplyDripMethod**. However,

CropAtRisk also cannot be concluded, because failure to prove ApplyDripMethod is not the same as proving ¬ApplyDripMethod.